Bacteriological contamination, dirt, and cracks of eggshells in furnished cages and noncage systems for laying hens: An international on-farm comparison

K. De Reu,*[1] T. B. Rodenburg,† K. Grijspeerdt,* W. Messens,* M. Heyndrickx,* F. A. M. Tuyttens,‡ B. Sonck,‡ J. Zoons,§ and L. Herman*

*Institute for Agricultural and Fisheries Research (ILVO), Technology and Food Science Unit, Brusselsesteenweg 370, 9090 Melle, Belgium; †Animal Breeding and Genomics Centre, Wageningen University, PO Box 338, 6700 AH Wageningen, the Netherlands; ‡Institute for Agricultural and Fisheries Research (ILVO), Animal Science Unit, Scheldeweg 68, 9090 Melle, Belgium; and §Provincial Centre for Applied Poultry Research, Poiel 77, 2440 Geel, Belgium

ABSTRACT For laying hens, the effects of housing system on bacterial eggshell contamination and eggshell quality is almost exclusively studied in experimental hen houses. The aim of this study was to compare eggshell hygiene and quality under commercial conditions. Six flocks of laying hens in furnished cages and 7 flocks in noncage systems were visited when hens were about 60 wk of age. Farms from Belgium, the Netherlands, and Germany were included in the study. The following parameters were determined on eggs sampled at the egg belts: 1) bacterial eggshell contamination, as expressed by total count of aerobic bacteria and number of Enterobacteriaceae; 2) proportion of dirty eggs; and 3) proportion of cracked eggs and eggs with microcracks. Considerable within-flock differences were found in eggshell contamination with total count of aerobic bacteria, both for furnished cages ($P \leq 0.001$, range 4.24 to 5.22 log cfu/eggshell) and noncage systems ($P \leq 0.001$, range 4.35 to 5.51 log cfu/eggshell). On average, lower levels of contamination with total count of aerobic bacteria (4.75 vs. 4.98 log cfu/eggshell; $P \leq 0.001$) were found on eggshells from furnished cages compared with noncage systems. Concerning Enterobacteriaceae, no significant difference in average eggshell contamination between both systems could be shown. The total percentage of cracked eggs was higher ($P \leq 0.01$) in furnished cages (7.8%) compared with noncage systems (4.1%). This was, however, due to the high percentage of cracked eggs (24%) observed on one of the furnished cage farms. We conclude that bacteriological eggshell contamination and percentage of cracked eggs differed substantially between individual farms using the same housing system. This may also explain some discrepancies between the findings of the present study versus some findings of previous experimental studies or studies on a small number of farms. Although statistically significant, the average differences in bacteriological contamination of nest eggs between both housing systems have limited microbiological relevancy.

Key words: laying hen, housing system, bacterial egg contamination, eggshell quality

2009 Poultry Science 88:2442–2448
doi:10.3382/ps.2009-00097

INTRODUCTION

Conventional cage housing for laying hens will be prohibited starting in 2012 in the European Union, following European Union Directive 1999/74 (Anonymous, 1999). From 2012 onward, only furnished cages and noncage systems (aviaries and floor housing) will be allowed. The alternatives for the conventional cages have been evaluated commercially, in terms of productivity and on bird welfare (Abrahamsson and Tauson, 1995; Tauson et al., 1999, 2002; Wall and Tauson, 2002; Wall et al., 2002; Rodenburg et al., 2005, 2008). In addition, attention was given to the effect of housing system on egg hygiene. The development toward furnished cages and noncage systems may have consequences for egg hygiene by increasing the percentage of cracked and dirty eggs (Wall and Tauson, 2002) or the bacterial eggshell contamination (De Reu et al., 2005b; Mallet et al., 2006). The effect of housing system on the bacterial eggshell contamination and eggshell quality has been studied almost exclusively in experimental hen houses. De Reu et al. (2006) suggested that the large differenc-

Received February 24, 2009.
Accepted August 4, 2009.
[1]Corresponding author: Koen.Dereu@ilvo.vlaanderen.be

es in bacteriological eggshell contamination found between conventional cages, furnished cages, and noncage housing systems in experimental studies might be less pronounced in commercial housing systems. The objective of our work was to analyze the hygiene (percentage of dirty eggs and eggshell bacterial contamination) and quality (cracks) of eggs produced in furnished cages and noncage systems under commercial conditions.

MATERIALS AND METHODS

Farms

Thirteen flocks of laying hens on 10 farms were included in this study (Table 1). Six flocks were housed in furnished cages and 7 flocks in noncage systems. Of these 7 flocks, 3 were housed in aviaries and 4 in floor housing systems. Only farms without an outdoor run were included. Regarding the farms with furnished cages, only farms were included with fully equipped furnished cages (perches, nest, scratching area). The farms were visited and sampled when the birds were about 60 wk of age. Farms from 3 different countries were included in the study (Belgium, the Netherlands, and Germany).

Number, Sampling, Collection, and Transport of Eggs

As described previously by De Reu et al. (2005a), sampling of 40 and 120 eggs was necessary to obtain statistically reliable results for the determination of the bacterial eggshell contamination and for the proportion of dirty, broken, hair-cracked, and microcracked eggs, respectively. On each farm, eggs were sampled at the egg belts next to the nests (no floor eggs). The eggs were sampled by hand (wearing clean gloves), placed in open carton filler flats, and transported by car, in ambient conditions, to the laboratory where they were kept for a maximum of 56 h in ambient conditions before analyzing (De Reu et al., 2005a).

Determination of Bacterial Eggshell Contamination

For the recuperation of bacteria from the eggshell, the intact egg was placed in a plastic bag with 10 mL of quarter-strength Ringer's solution (Oxoid, Hampshire, UK) and the egg was rubbed through the bag for 1 min (De Reu et al., 2005a). The diluent was plated on nutrient agar (CM 0003, Oxoid) for the determination of the total count of aerobic bacteria and on violet red bile glucose agar (356-4584, Bio-Rad, Hercules, CA) for the enumeration of Enterobacteriaceae. Plates were incubated, respectively, for 3 and 1 d at 30°C.

Bacteriological Air Contamination

An Air Sampler RCS (Biotest AG, Dreieich, Germany) was used to determine total count of aerobic bacteria and Enterobacteriaceae per cubic meter of air of the hen house. Strips in the air sampler contained nutrient agar and violet red bile glucose agar, respectively. Strips were incubated, respectively, for 3 and 1 d at 30°C.

Visual Examination of the Eggs

Each egg was thoroughly evaluated visually for the presence of dirt and placed into one or more of the following 3 categories: feces-blood, egg white-yolk, and dust-feathers (Anonymous, 1996). In addition, the occurrence of cracks—open cracks (visual), hair cracks (closed cracks), and microcracks (Bain, 2005)—in the eggshell was noted. The examination of the eggshell for the presence of hair cracks and microcracks was performed using a candling light.

Statistical Analysis of Data

The bacterial counts were log-transformed before statistical analysis (Jarvis, 1989). Significant differences were assessed using a 1-way ANOVA. The underlying assumptions for an ANOVA test were always verified:

Table 1. Overview of the visited flocks of laying hens per farm, system, country, the age of the birds (wk), and the hybrid used

Flock	Farm	Code	System	Country	Date	Age	Hybrid
1	1	FH1	Floor housing	the Netherlands	Sep. 2005	65	Hy-Line
2	2	FC1	Furnished cages	Belgium	Sep. 2005	60	Lohmann Selected Leghorn
3	3	FH2	Floor housing	the Netherlands	Oct. 2005	55	ISA Brown
4	2	FC2	Furnished cages	Belgium	Oct. 2005	62	ISA Brown
5	4	A1	Aviary system	Belgium	Nov. 2005	59	Bovans Goldline
6	5	A2	Aviary system	the Netherlands	Nov. 2005	59	Bovans Goldline
7	6	FC3	Furnished cages	Belgium	Jan. 2006	63	ISA Brown
8	6	A3	Aviary system	Belgium	Jan. 2006	63	ISA Warren
9	6	FC4	Furnished cages	Belgium	Jan. 2006	63	ISA Brown
10	7	FH3	Floor housing	Belgium	Jan. 2006	72	ISA Brown
11	8	FC5	Furnished cages	Germany	Feb. 2006	82	Lohmann Brown
12	9	FH4	Floor housing	Belgium	Feb. 2006	65	Bovans Goldline
13	10	FC6	Furnished cages	Germany	Jun. 2006	54	Lohmann Selected Leghorn

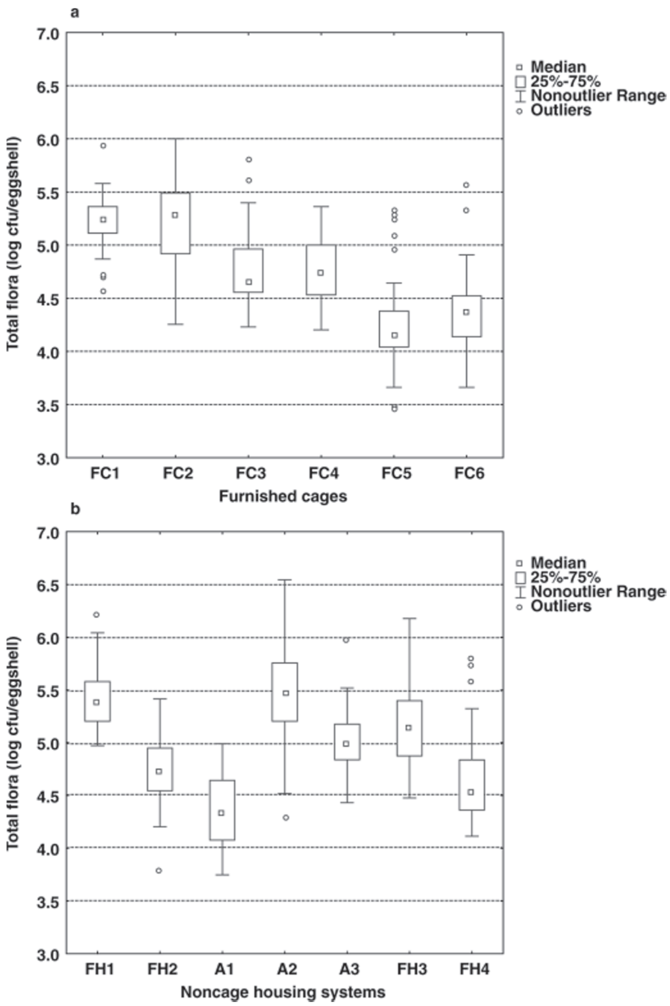

Figure 1. Eggshell contamination (n = 40 eggs) with total count of aerobic bacteria (total flora) in (a) the individual furnished cage flocks (FC) and (b) the individual noncage system flocks. FH = floor housing; A = aviary.

the homogeneity of variances and the absence of a correlation between means and variances was checked on a plot. Post hoc interfactor differences were calculated using Duncan's test (Kendall and Stewart, 1968). Interval estimation in the power analysis was used to calculate the confidence intervals at the 95% confidence level of the percentage cracks (open and hair cracks), microcracks, dirt, and type of dirt. A simple χ^2 test was used to assess whether the percentages differed between certain groups. A simple linear regression was carried out to determine the influence of air contamination on the bacterial eggshell contamination. The significance level α was set at 0.05. All analyses were done in Statistica 8.0 (Statsoft Inc., Tulsa, OK).

RESULTS

Bacteriological Eggshell Contamination

Figure 1a shows the individual results for the eggshell contamination with total count of aerobic bacteria for the different furnished cage flocks and Figure 1b for the different flocks from noncage systems. The figures show that both, within furnished cages and within noncage systems, major significant differences were obtained. Within the 6 furnished cages, differences in average eggshell contamination ranged between 4.24 and 5.22 log cfu/eggshell ($P \leq 0.001$). Comparable average differences were observed for the 7 noncage systems ($P \leq 0.001$, range 4.35 to 5.51 log cfu/eggshell).

A significant difference ($P \leq 0.001$) was found between the average eggshell contamination of eggs from all furnished cages (n = 230 eggs) compared with all noncage systems (n = 278 eggs; Figure 2a). Eggshells from furnished cages were on average less contaminated with total count of aerobic bacteria compared with noncage eggshells (4.75 vs. 4.98 log cfu/eggshell).

On the other hand, within the noncage systems, the average eggshell contamination with total count of aerobic bacteria found in the 4 floor housing systems (5.00 log cfu/eggshell; n = 158 eggs) was not significantly different ($P > 0.05$) from the average contamination in the 3 aviary systems (4.95 log cfu/eggshell; n = 120 eggs; Figure 2b).

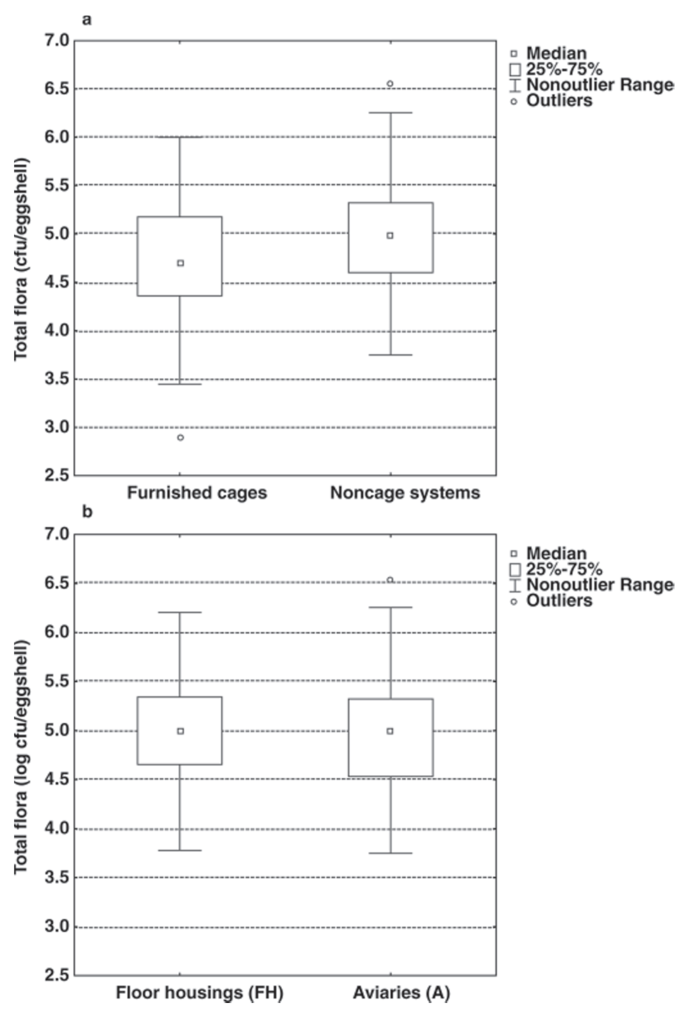

Figure 2. Eggshell contamination with total count of aerobic bacteria in (a) the noncage systems compared with the furnished cages and in (b) the 2 types of noncage housings systems.

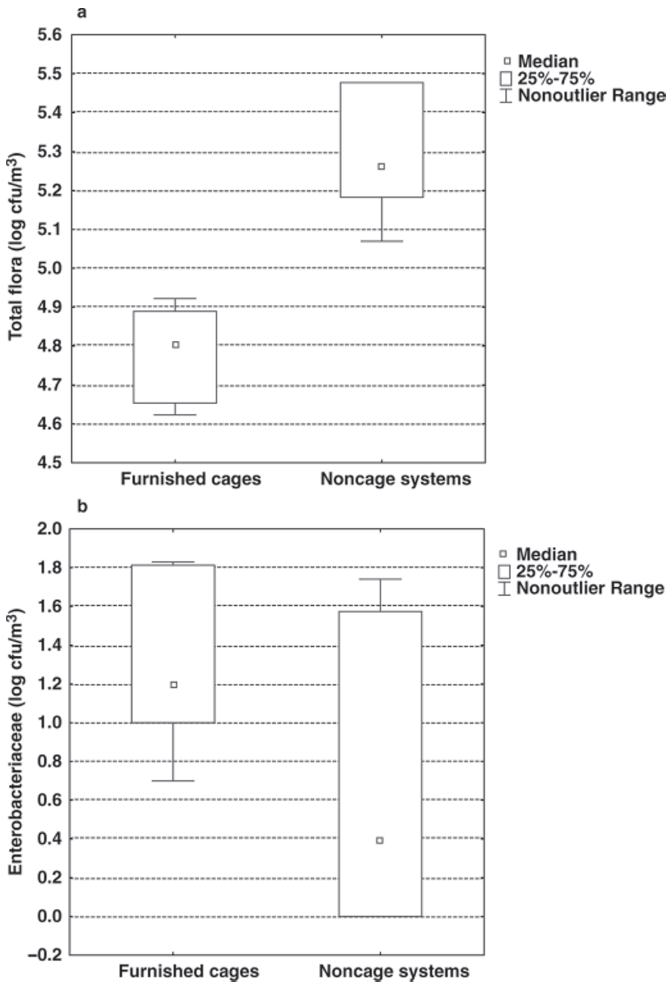

Figure 3. Bacteriological air contamination with (a) total count of aerobic bacteria and (b) number of Enterobacteriaceae in the noncage systems compared with the furnished cages.

For Enterobacteriaceae, no significant ($P > 0.05$) difference in average eggshell contamination was found between furnished cages and noncage systems. Respectively, 88 and 94% of eggshells contained <10 cfu Enterobacteriaceae/eggshell (Table 2). The remaining 12% of the 230 eggs from furnished cages and 6% of the 278 eggs from the noncage systems had countable Enterobacteriaceae results (≥10 cfu). Averages of those counts were 1.51 ± 0.63 and 1.54 ± 0.76 log cfu/eggshell, respectively.

Bacteriological Air Contamination

The average bacteriological contamination of the air with total count of aerobic bacteria was significantly higher ($P \leq 0.001$) for noncage systems (5.31 log cfu/m^3) compared with furnished cages (4.75 log cfu/m^3; Figure 3a). No significant ($P > 0.05$) difference was found between floor housing and aviary housing (5.26 vs. 5.38 log cfu/m^3 of air). On the other hand, air contamination with Enterobacteriaceae was lower, although not significantly ($P > 0.05$), in noncage systems (0.72 log cfu/m^3) compared with furnished cages (1.35 log cfu/m^3; Figure 3b). In addition, no significant ($P >$ 0.05) difference was found between floor housing and aviary housing (0.83 vs. 0.58 log cfu/m^3 of air).

Visual Examination of the Eggs

The total percentage of cracked and hair-cracked eggs was higher ($P \leq 0.01$) in the furnished cages (7.8%) compared with the noncage systems (4.1%; Table 3). This was due to the high percentage (24%) observed on one of the furnished cage farms (flock 7). Elimination of this single deviating result leads to no significant difference ($P > 0.05$) in the percentage of cracks for furnished cages (4.5%) compared with noncage systems (4.1%). No significant difference in percentage of dirty eggs (overall dirt) was found between furnished cages and noncage systems (Table 3). Within the noncage systems, a significant difference was found for broken and hair-cracked eggs between aviaries and floor housing ($P \leq 0.05$), whereas no difference in dirty eggs was found between aviaries and floor housing.

DISCUSSION

Comparing the contamination of eggs laid in different types of experimental housing systems, Protais et al. (2003b) and De Reu et al. (2005b) also found higher eggshell contamination with total count of aerobic bacteria in aviaries compared with cage systems (conventional and furnished cages). The increase found in these studies was more than 1 log cfu (up to a total of 6.0 log cfu/shell; ignoring floor eggs), compared with an average difference of only 0.23 log cfu between furnished cages and noncage systems in the present study on multiple farms. Although the difference of 0.23 log

Table 2. Counts of Enterobacteriaceae on the eggshell for the different housing systems

			>10 cfu/eggshell	
Item	n	<10 cfu/eggshell (%)	Average log	SD log
Furnished cages	230	88	1.51	0.63
Floor housing	158	93	1.62	0.88
Aviary	120	96	1.37	0.41
Noncage housing	278	94	1.54	0.76

cfu is statistically significant ($P \leq 0.001$), in microbiology, the difference is of limited relevancy. In a study in which the contamination progress in the egg handling chain was followed, De Reu et al. (2006) already found a smaller difference between commercial cage and noncage systems compared with their experimental study. On average, a higher initial eggshell contamination with total count of aerobic bacteria was found for eggs from both noncage systems (5.46 log cfu/eggshell) compared with both conventional cage systems (5.08 log cfu/eggshell; De Reu et al., 2006). On the other hand, in the study of Schwarz et al. (1999), in which sampling was performed in a packing station, free-range eggs were also about 1 log higher contaminated with aerobic bacteria compared with conventional cage eggs.

Notwithstanding the significant difference in eggshell contamination with total count of aerobic bacteria, for Enterobacteriaceae, no significant difference in average eggshell contamination was found between furnished cages and noncage systems. In the experimental study of De Reu et al. (2005b), also no systematic differences for gram-negative bacteria were found between conventional cages, furnished cages, and aviary systems. Schwarz et al. (1999) even found a higher percentage of gram-negative bacteria on cage eggs compared with free-range eggs. Similarly, in the study of the contamination progress in the egg handling chain, contamination with gram-negative bacteria was also lower in the commercial noncage systems (3.31 log cfu/eggshell) compared with the 2 conventional cages (3.85 log cfu/eggshell; De Reu et al., 2006). Wall et al. (2008), comparing conventional and furnished cages in an experimental study, found Enterobacteriaceae on a significantly higher portion of eggs in furnished cages (12.3% on average) than in conventional cages (5.8% on average). In the present study, we found Enterobacteriaceae on 12% of the furnished cage eggs and only on 6% of the noncage eggs. The tendency for a lower contamination with gram-negative bacteria in general and Enterobacteriaceae specifically on noncage eggs compared with cage eggs is remarkable. We assume that the higher amount of gram-positive bacteria on the shell of noncage eggs possibly suppresses the presence of gram-negative bacteria and Enterobacteriaceae.

Comparing the average eggshell contamination between the 4 floor housing flocks and the 3 aviary flocks, no significant differences were obtained for total count of aerobic bacteria nor for Enterobacteriaceae. In the previously mentioned study of De Reu et al. (2006) comparing a floor housing system with an aviary system, the difference in eggshell contamination with total count of aerobic bacteria was 0.2 log cfu/eggshell and >0.2 log for gram-negative bacteria.

The observed significant differences in total count of aerobic bacteria within the different farms with the same type of housing system are remarkable (>1 log cfu/eggshell). These results indicate that not only housing system but also differences in farm construction or management play an important role in the bacterial eggshell contamination. The important role of farm management is confirmed by the results obtained on farm 6 where both housing systems were available and only a difference of 0.3 log cfu was found between both furnished cage flocks (FC3 and FC4) and the aviary flock (A3). In the study of De Reu et al. (2006), comparing 2 conventional cage systems, 1 aviary and 1 floor housing system, from 4 different farms, the differences found within the same housing system were also smaller (0.2 log cfu/eggshell between both cage systems as well between both noncage systems). The results of our international on-farm study show that comparisons of housing systems must be based on a relatively large number of farms and flocks. Our study also explains some differences in results found between experimental studies using 1 farm with 1 management and other studies based on multiple farms with presumably different management.

The total count of aerobic bacteria in the air of the experimental poultry houses has been found to be positively correlated with the bacterial eggshell contamination at the hen house (Protais et al., 2003a; De Reu et al., 2005b). Averages of 4 log cfu/m^3 of air for the conventional and furnished cages were found compared with a 100 times higher average (>6 log cfu/m^3) for

Table 3. Occurrence of cracks, microcracks, and dirt in or on the eggshell for the different housing systems[1]

Item	n	Cracks (%)	Microcracks (%)	Feces-blood (%)	Egg white-egg yolk (%)	Dust-feathers (%)	Overall dirt (%)
Furnished cages	717	7.8^{AA}	2.8	8.9	1.9^{AAA}	13.5^{AAA}	22.0
		[5.7; 10.4]	[1.8; 4.3]	[6.7; 11.5]	[0.9; 3.4]	[10.6; 16.6]	[18.6; 25.6]
Floor housing	420	5.7^{a}	1.9	10.7	11.7^{aaa}	5.4^{a}	26.2
		[3.7; 8.4]	[0.8; 3.7]	[7.9; 14.1]	[8.8; 15.1]	[3.5; 8.1]	[22.0; 30.7]
Aviary	358	2.2^{b}	1.1	10.3	3.4^{bbb}	9.4^{b}	21.5
		[1.0; 4.4]	[0.3; 2.8]	[7.4; 14.0]	[1.7; 5.8]	[6.7; 13.0]	[17.4; 26.1]
Noncage housing	778	4.1^{BB}	1.5	10.5	7.8^{BBB}	7.3^{BBB}	24.0
		[2.8; 5.8]	[0.8; 2.7]	[8.5; 12.9]	[6.0; 10.0]	[5.6; 9.4]	[21.1; 27.2]

^{a,b}Numbers in the same column with different lowercase superscripts show significant differences between the noncage housing systems, floor housing, and aviary ($^{a,b}P \leq 0.05$; $^{aaa,bbb}P \leq 0.001$).

^{A,B}Numbers in the same column with different uppercase superscripts show significant differences between furnished cages and noncage housing ($^{AA,BB}P \leq 0.01$; $^{AAA,BBB}P \leq 0.001$).

[1][x; y]: x = lower confidence limit in percentage; y = upper confidence limit in percentage; both at 95% confidence level.

aviary housing systems. In the experimental study of De Reu et al. (2005b), a positive correlation was found between the concentration of aerobic bacteria in the air of the poultry houses and the eggshell contamination. Differences in air contamination with total count of aerobic bacteria found in our present extensive on-farm comparison were also significant ($P \leq 0.001$) but much smaller compared with the experimental studies. No significant ($P > 0.05$) difference was found between floor housing and aviary housing. For Enterobacteriaceae, no significant difference ($P > 0.05$) in air contamination between furnished cage and noncage and between floor and aviary housing was found. In contrast to the experimental study, in this multiple on-farm comparison, no significant correlation was found between the concentration of aerobic bacteria in the air of the poultry houses and the eggshell contamination. This confirms the findings of De Reu et al. (2006) comparing 2 commercial conventional cages and 2 noncage housing systems, in which only a moderate but not significant positive correlation was found.

Although significant differences were found within the individual visual categories of dirt between furnished cages and noncage systems, the overall percentage of eggs with dirt on the shell was not significantly different between both housing systems. Comparing the floor housing systems with the aviary systems, also no significant difference in overall percentage of eggs with dirt on the shell was found (26.2 vs. 21.5%). Tauson et al. (1999) compared 3 housing systems in the same experimental building (conventional cages, an aviary system, and a floor housing system) and 2 genotypes of hens (Lohmann Selected Leghorn and Lohmann Brown). Eggshell dirtiness from eggs collected from the belts did not differ significantly between housing systems for Lohmann Brown, but for Lohmann Selected Leghorn, dirty eggs were fewer in the floor system than in either the aviary or the cages. The percentage of dirty eggs varied from 4.0 to 8.8%.

Our study documented variations of cracked eggs from 0 to 24% for the individual farms (data not shown). The high percentage of cracks in one flock showed again that not only housing systems but also farm construction and management are important and that comparison of housing systems must be based on an adequate number of farms. Guesdon et al. (2006) found 15.4 to 19.6% of broken (visual observation) and hair-cracked (candling) eggs in furnished cages compared with only 8.1 to 12.2% in standard cages (experimental studies). This was mainly due to hair-cracked eggs in the narrow nests of the furnished cages without egg saver and the relatively low frequency of manual egg collection. The high percentage (24%) of cracks we found in the furnished cage flock 7 was probably caused by a bad adjustment of the egg saver and the accumulation of eggs next to the nest box on a short part of the egg belt. In the study of Tauson et al. (1999), cracks varied from 2.2 to 7.7%, with no significant difference between the conventional, furnished, and floor housing systems. In a market study on the quality characteristics of eggs from different housing systems, Hidalgo et al. (2008) found no significant difference ($P > 0.05$) in appearance of cracked eggs between cage (14%), free-range (10%), barn (11%), and organic (5%) eggs.

We conclude that the bacteriological contamination of eggs and the eggshell quality differed substantially between individual farms using the same housing system. This may also explain some discrepancies between the findings of the present study versus some findings of previous experimental studies on a very small number of farms. We found that average differences in bacteriological contamination of air, eggs, and eggshell quality between the furnished cage and noncage systems were rather limited for eggs sampled at the egg belts, whereas differences between floor housing and aviary are negligible.

ACKNOWLEDGMENTS

We thank all of the farmers who were willing to participate in this study. This project was funded by the Department of Health, Food Chain Safety and Environment of the federal government of Belgium. The practical work was performed especially by Ann Van de Walle, but Jürgen Baert, Willy Bracke, and Vera Van de Mergel, all personnel of ILVO, Melle, Belgium, are also gratefully acknowledged.

REFERENCES

Abrahamsson, P., and R. Tauson. 1995. Aviary systems and conventional cages for laying hens. Acta. Agric. Scand. 45:191–203.

Anonymous. 1996. Begripsomschrijving tweede soort eieren. Pages 1–15 in Product Boards for Livestock, Meat and Eggs, Rijswijk, the Netherlands.

Anonymous. 1999. Council Directive 1999/74/EC on 19 July 1999 laying down minimum standards for the protection of laying hens. Off. J. Eur. Communities L203:53–57.

Bain, M. 2005. Recent advances in the assessment of eggshell quality and their future application. World's Poult. Sci. J. 61:268–277.

De Reu, K., K. Grijspeerdt, M. Heyndrickx, M. Uyttendaele, J. Debevere, and L. Herman. 2006. Bacterial eggshell contamination in the egg collection chains of different housing systems. Br. Poult. Sci. 47:163–172.

De Reu, K., K. Grijspeerdt, M. Heyndrickx, M. Uyttendaele, and L. Herman. 2005a. The use of total aerobic and Gram-negative flora for quality assurance in the production chain of consumption eggs. Food Contr. 16:147–155.

De Reu, K., K. Grijspeerdt, M. Heyndrickx, J. Zoons, K. De Baere, M. Uyttendaele, J. Debevere, and L. Herman. 2005b. Bacterial eggshell contamination in conventional cages, furnished cages and aviary housing systems. Br. Poult. Sci. 46:149–155.

Guesdon, V., A. M. H. Ahmed, S. Mallet, J. M. Faure, and Y. Nys. 2006. Effects of beak trimming and cage design on laying hen performance and egg quality. Br. Poult. Sci. 47:1–12.

Hidalgo, A., M. Rossi, F. Clerici, and S. Ratti. 2008. A market study on the quality characteristics of eggs from different housing systems. Food Chem. 106:1301–1308.

Jarvis, B. 1989. Statistical Aspects of the Microbiological Analysis of Foods. Progress in Industrial Microbiology. M. E. Bushell and B. Jarvis, ed. Elsevier, New York, NY.

Kendall, M. G., and A. Stewart. 1968. Pages 1–557 in The Advanced Theory of Statistics. Volume 3: Design and Analysis, and Time-Experiments. M. G. Kendall and A. Stewart, ed. Charles Griffin & Co. Ltd., London, UK.

Mallet, S., V. Guesdon, A. M. H. Ahmed, and Y. Nys. 2006. Comparison of eggshell hygiene in two housing systems: Standard and furnished cages. Br. Poult. Sci. 47:30–35.

Protais, J., S. Quesguiner, E. Boscher, J.-C. Piquet, B. Nagard, and G. Salvat. 2003a. Effect of housing system on the bacterial flora in the air and on egg shells. Pages 142–149 in Proc. 10th European Symposium on the Quality of Eggs and Egg Products, St. Brieuc-Ploufragan, France. G. Salvat and J. Protais, ed. St-Brieuc, France.

Protais, J., S. Quesguiner, E. Boscher, J.-C. Piquet, B. Nagard, and G. Salvat. 2003b. Effect of housing system on the bacterial flora of egg shells. Br. Poult. Sci. 44:788–790.

Rodenburg, T. B., F. Tuyttens, K. De Reu, L. Herman, J. Zoons, and B. Sonck. 2005. Welfare, health and hygiene of laying hens housed in furnished cages and in alternative housing systems. J. Appl. Anim. Welf. Sci. 8:211–226.

Rodenburg, B., F. Tuyttens, K. De Reu, L. Herman, J. Zoons, and B. Sonck. 2008. Welfare assessment of laying hens in furnished cages and non-cage systems: An on-farm comparison. Anim. Welf. 17:363–373.

Schwarz, G., A. Kobe, and R. Fries. 1999. Microflora on egg shells from different housing systems. Arch. Geflügelkd. 5:220–224.

Tauson, R. 2002. Furnished cages and aviaries: Production and health. World's Poult. Sci. J. 58:49–63.

Tauson, R., A. Wahlström, and P. Abrahamsson. 1999. Effect of two floor housing systems and cages on health, production, and fear response in layers. J. Appl. Poult. Res. 8:152–159.

Wall, H., and R. Tauson. 2002. Egg quality in furnished cages for laying hens—Effects of cracked reduction measures and hybrid. Poult. Sci. 81:340–348.

Wall, H., R. Tauson, and K. Elwinger. 2002. Effects of nest design, passages and hybrid on use of nest and production performance of layers in furnished cages. Poult. Sci. 81:333–339.

Wall, H., R. Tauson, and S. Sorgjerd. 2008. Bacterial contamination of eggshells in furnished and conventional cages. J. Appl. Poult. Res. 17:11–16.